

Topic: Testing, Debugging, Exceptions, Assertions

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① Check My Understanding

Questions completed: 5 / 5

Answers revised after AI hints: 5 / 5

② My Misconception

Before: I thought...

I thought that if a program passed all the tests, it meant that the program had no bugs.

Now: I understand...

Passing tests only means the program works correctly for the cases that were tested. There may still be bugs in situations that the tests did not cover. I also learned that debugging is the process of finding and fixing the cause of a problem, while exceptions and assertions can help detect problems and make errors easier to identify.

③ Challenge the AI

One AI-generated question I challenged:

Assertions should be used to handle errors that may normally happen when a program is running.

Why?

☐ Ambiguous

☐ Oversimplified

☒ Technically questionable

☐ Too easy

☐ Other: _____

Brief explanation:

I think this statement is technically questionable because assertions are mainly used to check conditions that the programmer expects to be true. They are useful for finding programming mistakes during development, but they are not usually the best way to handle normal situations such as invalid user input. Exceptions are more appropriate for handling errors that may normally occur during program execution.

④ One-Minute Reflection

One thing I am still unsure about:

I am still unsure about when I should use an assertion and when I should use an exception. I understand their basic purposes, but sometimes it is difficult to decide which one is more appropriate in a real program.